

DO NOT WRITE ON TEST BOOKLET



**FINANCIAL MATH &
ANALYSIS CONCEPTS
(190)**

—OPEN EVENT—

STATE – 2020

TOTAL POINTS

_____ ***(100 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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Directions: Answer the following multiple-choice questions by identifying the letter that best completes each sentence or answers the question.

1. David is going to transfer a \$2,225 balance from one credit card to a new credit card. The new credit card charges a 2.5% fee for balance transfers. What fee will David pay to transfer the balance to the new card?
 - a. \$55.63
 - b. \$56.55
 - c. \$53.65
 - d. \$55.36

2. Lexi borrowed \$5,550 from her local bank to buy a used car. Lexi signed a 2-year promissory note at 12% interest. Find the amount of interest Lexi will pay when the note is due.
 - a. \$1,233.00
 - b. \$1,333.00
 - c. \$1,332.00
 - d. \$1,232.00

3. Mary is considering updating her barn into a wedding venue. She decides to do some initial calculations to see how much of an investment she will be making. She calculates the overall cost to be \$78,450.00. However, her accountant tells her that she will receive a 30% return on her investment through her upcoming tax return. Therefore, how much is the initial cost to Mary?
 - a. \$59,415.00
 - b. \$54,915.00
 - c. \$59,451.00
 - d. \$54,951.00

4. Natalie signed a promissory note for \$9,500 at 6% for 1 year. What is the total due on the promissory note when it matures?
 - a. \$ 9,500.00
 - b. \$ 570.00
 - c. \$ 8,930.00
 - d. \$10,070.00

5. Kevin is an administrative assistant to the CEO of his company. His annual salary is \$87,950.00. How much is his gross pay if he is paid bi-weekly?
 - a. \$3,664.58
 - b. \$7,329.16
 - c. \$3,382.69
 - d. \$3,328.96

6. Jack's Donuts bakes 1,652 donuts every morning for the morning rush. Their sales on Monday morning were \$2,482.00. What is the average price per donut sold?
 - a. \$1.50
 - b. \$1.25
 - c. \$0.95
 - d. \$1.00

7. You decide to purchase a new laptop computer through an installment loan company. The cash price for the laptop is \$1,300. To purchase the computer on an installment loan, you must furnish a 10% down payment. If your installment loan payments are \$48.00 for 36 months, what is the finance charge total you will pay to purchase?
 - a. \$558.00
 - b. \$585.00
 - c. \$855.00
 - d. \$885.00

8. Ellie works as a medical assistant at a local hospital. One of her main jobs is medical billing for all patients seen by the emergency room. Ellie worked this week for a total of 44 hours at \$22.00 per hour. All hours over 40 are considered time-and-half. In these 44 hours she filed 19,750 claims. What is the average cost per claim given this information?
 - a. \$0.08 per claim
 - b. \$0.04 per claim
 - c. \$0.05 per claim
 - d. \$0.07 per claim

9. An online purchase included: 1 shirt @ \$18.99, 1 sweater @ \$22.99, 6 pairs of socks @ \$5.99 each, and 1 jacket @ \$145.99. Sales tax in your state is 9.5%. Shipping costs are 13% of the overall sale prior to tax. What is the total cost of the online purchase?
 - a. \$274.29
 - b. \$267.59
 - c. \$268.36
 - d. \$286.69

10. Charles had total earnings of \$143,789 last year while living in California. He paid a total of \$44,600 to rent his home where he lived. What percentage of Charles' overall earnings did rent represent?
 - a. 30.0%
 - b. 31.0%
 - c. 31.7%
 - d. 31.8%

11. Jaden works as a furniture representative for a top competing furniture company. His sales for the last quarter were \$452,789. He is paid a graduated commission. His commission rates are: first 10,000 is 5%, the next 20,000 is 6.5%, the next 70,000 is 9%, and anything over 100,000 in sales pays 10%. If Jaden continues to have the same average sales for the entire year, how much will Jaden earn as a gross income?
 - a. \$ 43,378.90
 - b. \$ 86,757.80
 - c. \$173,515.60
 - d. \$175,315.80

12. The Hose Shop, a local retailer marketing different types of garden hoses had a total profit of \$123,658 last year. If the total revenue dollars was \$1,456,299. What is the market share for the local retailer?
 - a. 8.4%
 - b. 8.25%
 - c. 8.48%
 - d. 8.5%

13. PJ works at a button factory where he makes buttons for various garments. Last week, PJ's production was: 72, 86, 44, 89, and 101. PJ earns \$1.45 per button he creates on top of his salary of \$400.00 for the week. How much did PJ earn this past week in gross earnings?
 - a. \$568.40
 - b. \$586.40
 - c. \$968.40
 - d. \$698.40

14. Megan takes her client out to dinner during a business meeting while in town. Their bill totaled \$95.32. She wants to tip industry standard of 20% for their server. What is the total bill that Megan paid for the meal?
 - a. \$ 95.32
 - b. \$ 97.23
 - c. \$114.37
 - d. \$114.38

15. Rockets has an invoice totaling \$423,222 and paid it before their discount period expired. Their total payment on their check was \$412,222. What was Rockets discount rate with this accounts payable?
 - a. 2.60%
 - b. 2.62%
 - c. 2.75%
 - d. 2.25%

16. Jackson earned \$6,428 working part-time while he was a student this past year. He wanted to have 35% of his annual income in his savings account. How much would he have to save from his bi-weekly payroll check to have 35% of his annual income in his savings account?
 - a. \$86.53
 - b. \$85.63
 - c. \$84.25
 - d. \$82.45

17. Jason is completing his mileage on his expense report for his company. He earns \$0.30 per mile over 10 miles per trip. His recent trips were: 22, 30, and 45 miles. How much reimbursement can Jason expect on this expense report for mileage?
 - a. \$29.10
 - b. \$29.01
 - c. \$20.01
 - d. \$20.10

18. Eddie insured his home for \$45,000 at an annual rate of \$0.52 per \$100. What is Eddie's premium?
 - a. \$ 234.00
 - b. \$ 243.00
 - c. \$ 285.00
 - d. \$45,000.00

19. After taking inventory of your home including all valuables, you decide your total asset value is \$95,500.00. If the total insurance premium is \$783.10 and is based off of a rate per \$100 assessed, what is the rate per \$100?
 - a. \$0.76
 - b. \$0.69
 - c. \$0.82
 - d. \$0.89

20. Roger is a renter and just moved into a new apartment. His total square footage is 1,201 in his new 2 bedroom, 2 bath apartment. He pays an insurance premium of \$0.15 per square foot of his apartment. What is total premium?
 - a. \$180.15
 - b. \$168.45
 - c. \$181.50
 - d. \$169.15

21. Adam had a house fire last month. His property was insured for \$80,000 with a \$1,000 deductible. How much will the insurance company pay for his home which was a total loss?
 - a. \$80,000
 - b. \$79,000
 - c. \$81,000
 - d. \$78,000

22. Dolly's Doodles Depot an art gallery has \$745,200 in assets. The company's equity is \$715,000. How much liability does Dolly's carry?
 - a. \$30,000
 - b. \$31,000
 - c. \$32,200
 - d. \$30,200

23. What is the Manufacturer Suggested Retail Price (MSRP) for a car on a dealership lot that has a base price of \$28,328 and \$1,345 in additional features?
 - a. \$26,983
 - b. \$28,328
 - c. \$29,673
 - d. \$29,763

24. Dan's yearly salary is \$45,500. His monthly fixed expenses include rent and car payments total \$650.00 a month. His car payment is 40% of his fixed expenses for the month. What percentage does his car payment make up his overall budget for the year?
 - a. 6.9%
 - b. 6.7%
 - c. 6.6%
 - d. 6.5%

25. Steve's purchase price for a new car was 85% of the \$23,500 MSRP. Sales tax was figured at 7%. Registration fees totaled \$272. He received a customer discount rebate of \$600. His trade-in valued at \$2,000 was his only down payment. What is the balance due on the car?
 - a. \$19,045.25
 - b. \$22,817.00
 - c. \$18,773.25
 - d. \$22,678.25

26. Connie's credit card has an APR of 13% and uses a monthly periodic rate. Her balance subject to finance charge is \$413.85. How much is the finance charge for this month's billing cycle?
 - a. \$4.45
 - b. \$4.54
 - c. \$4.48
 - d. \$4.47

27. Louise purchased a car for \$13,750. She has estimated her trade-in value to be \$3,225 at the end of 6 years. Find the estimated total average annual depreciation of the car.
- \$10,525.00
 - \$ 1,754.17
 - \$ 933.33
 - \$10,333.33
28. Delrosa Melendez has a mortgage with the 1st National Trust Bank. The bank requires that she pays her homeowner's insurance, property taxes, and mortgage in one monthly payment. Their monthly mortgage payment is \$1,722, their semi-annual property tax bill is \$6,110, and their quarterly homeowner's insurance bill is \$457. How much does Delrosa pay the bank each month?
- \$2,892.66
 - \$ 589.42
 - \$2,740.33
 - \$1,836.25
29. Kathleen is self-employed and received a 1099 MISC tax form for her annual income earned as a contractor in the amount of \$12,650.00. If the Social Security and Medicare rate is 7.65% combined, and she must pay the portion that a company usually matches, how much is her self-employment taxes?
- \$ 967.73
 - \$ 976.73
 - \$1,935.45
 - \$1,945.35
30. Samantha had four x-rays taken at the total cost of \$275. Under her major medical insurance coverage, her insurance company paid 80% of the cost of the x-rays after she paid her \$35 deductible fee for each x-ray. What was the insurance company's share of the cost of the x-rays?
- \$275.00
 - \$220.00
 - \$180.00
 - \$108.00
31. What will Samantha's share be when the bill is due if Samantha had four x-rays taken at the total cost of \$275. Under her major medical insurance coverage, her insurance company paid 80% of the cost of the x-rays after she paid her \$35 deductible fee for each x-ray.
- \$167.00
 - \$ 55.00
 - \$ 0.00
 - \$ 95.00

32. If the hospital were to give Samantha a 12% discount on her portion of payment due for paying within the 15-day grace period; what is Samantha's new payment if her original amount due was \$167.00?
- \$ 13.92
 - \$146.96
 - \$164.96
 - \$ 11.13
33. Mr. Fix-It Screen Inc., developed cellphone non-shattering protector screens for civilian use. In a test market, Mr. Fix-It sold 9,375 units at \$19.99 per unit. Mr. Fix-It wants to sell the units globally in all 125 markets they currently sell. Based on sales in the test market, forecast future sales in all 125 markets.
- \$ 187,406.25
 - \$ 183,406.25
 - \$23,425,781.25
 - \$23,452,781.25
34. You are updating your home office and need two file cabinets to complete the redesign. You noticed that the local office superstore offers the file cabinets you are looking for at \$149.99 each. They are running a special discount for self-employed employees. The discount is 5% when you buy two of any matching furniture or office equipment. What is the total bill for your two file cabinets?
- \$284.98
 - \$285.00
 - \$284.99
 - \$285.98
35. Annie earned a monthly salary of \$1,850.00. She budgets her entertainment at 22% of her monthly salary. What is her budget amount for entertainment?
- \$370.00
 - \$403.00
 - \$405.00
 - \$407.00
36. Michael earns 12 sick days per year that he works at his place of employment. He also earns 4 personal days that can be converted to sick days at the end of the year. Last year, Michael banked 245 sick days. This year he used 4 sick days, and 2 personal days. At the end of the year, how many sick days will Michael have in reserve?
- 245 sick days
 - 240 sick days
 - 255 sick days
 - 257 sick days

37. Suzanne invests in a new company. She purchases 1,000 shares at \$6.50 per share. Her broker charges a fee of 3.25%. What is the total cost of her investment?
- \$6,500.00
 - \$6,000.00
 - \$6,711.25
 - \$8,612.50
38. K. Kotyuk buys a \$500 bond at 105.874. The broker charges \$4 commission per bond with a minimum of \$40 commission per order. What commission was Kevin Kotyuk charged?
- \$40
 - \$28
 - \$32
 - \$29
39. K. Kotyuk buys a \$500 bond at 105.874. The broker charges \$4 commission per bond with a minimum of \$40 commission per order. What was the total amount he invested in the bond?
- \$569.73
 - \$596.73
 - \$566.93
 - \$569.37
40. K. Kotyuk buys a \$500 bond at 105.874. The broker charges \$4 commission per bond with a minimum of \$40 commission per order. What is the amount of premium he paid on the bond?
- \$69.73
 - \$69.37
 - \$96.73
 - \$66.93
41. Tamia earns \$18,900 for the year at a local retail store. What is Tamia's hourly wage if she works a standard 40 hour work week every week of the year?
- \$9.09
 - \$9.10
 - \$9.11
 - \$9.07
42. BuckStars Coffee paid Suzie a total of \$1,678 in dividends last year. Suzie owns 352 shares of BuckStars. What was the semiannually dividend for one share of BuckStars?
- \$2.38
 - \$2.39
 - \$4.76
 - \$4.77

43. The Small Township School Corporation is selling bonds at 108.489. What is the market price of one of their \$1,000 bonds? Are they selling for a discount or premium?
- \$1,084.89, discount
 - \$1,084.89, premium
 - \$108.49, premium
 - \$108.49, discount
44. The Yellow Bus Company offered \$1,000, 6% bonds at 101.585. What is the current yield, to the nearest tenth percent?
- 5.8%
 - 6.0%
 - 5.9%
 - 5.5%
45. Wendy earns \$14.50 an hour. This week she worked a total of 45 hours. All hours over 40 are paid at time-and-a-half. Her taxes were \$102.85 for the pay period. What is her net pay?
- \$688.75
 - \$584.90
 - \$585.90
 - \$687.75
46. Burgers Limited executed a 2-for-1-split in May. Joseph had 940 shares before the split. Each share was worth \$74.40. How many shares will Joseph have after the split and what is their value?
- 1,880 shares, \$37.20
 - 470 shares, \$148.80
 - 470 shares, \$37.20
 - 1,880 shares, \$148.80
47. Elias received his bank statement. The statement shows a balance of \$593.22. Checks outstanding are: #941 for \$29.99 and #942 for \$39.33. A \$125 deposit was outstanding. Reconcile Elias's bank statement balance.
- \$523.90
 - \$648.90
 - \$718.22
 - \$648.22
48. Bob invests 23% of his annual income to his savings account. His annual salary is \$37,800.00. What is his total investment for the year?
- \$8,694.00
 - \$8,964.00
 - \$8,694.50
 - \$8,964.50

49. Johnny pays \$64.00 bi-weekly for health insurance. What is Johnny's annual healthcare cost?
- a. \$1,664
 - b. \$1,536
 - c. \$3,328
 - d. \$ 768
50. Bonnie loves her new dog. She recently inquired about pet insurance on her dog Sparky. If the annual insurance premium is \$765.00, how much is her weekly insurance cost for her budget?
- a. \$14.71
 - b. \$63.75
 - c. \$15.93
 - d. \$15.94